Starter

The PowerPoint or interactive could be used as a quick labelling activity to introduce the modelling activity.

Modelling the circulatory system

Depending on the group the students could be asked to design their own simulation using the following resources.

Resources

- 2 large hoops
- 100 red and blue cards
- 2 boxes/bucket
- labels for heart, lungs and cells

Task

Assign students to a role: Lungs (1 person), cells (1 person), heart (2 people to hold hoops), blood (rest of class or a smaller group).

Arrange students around the room (refer to animation on slide 4).
Students move around to represent blood flow – picking up red cards at the lungs; going through the hoop (heart); dropping off the red cards at the cells and picking up blue cards; returning back through the heart (hoop) to lungs and dropping of blue cards.
The circulation will stop when all the blue cards are at the lungs and the red cards are at the cells.

Discussion

There are many difficulties with this model. Get the students to recognise some of them and suggest how the model could be improved. The following questions could be asked to stimulate discussion.

Did the model show how oxygen and carbon dioxide are carried in the blood? – oxygen in red blood cells, carbon dioxide dissolved in plasma.

In the model circulation stopped, why does this not happen in reality? – discuss that concentration gradients are maintained in lungs due to gas exchange and blood flow and in cells due to respiration.

Was the movement of the blood realistic in the model? – discuss network of blood vessels.

Ask students to draw the model and evaluate it.